# Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

**Q.1**

Which of the following will NOT be observed when a multimeter (operating in resistance measuring mode) probes connected across a component, are just reversed?

**(A)** Multimeter shows an equal deflection in both cases i.e., before and after reversing the probes if the chosen component is resistor.

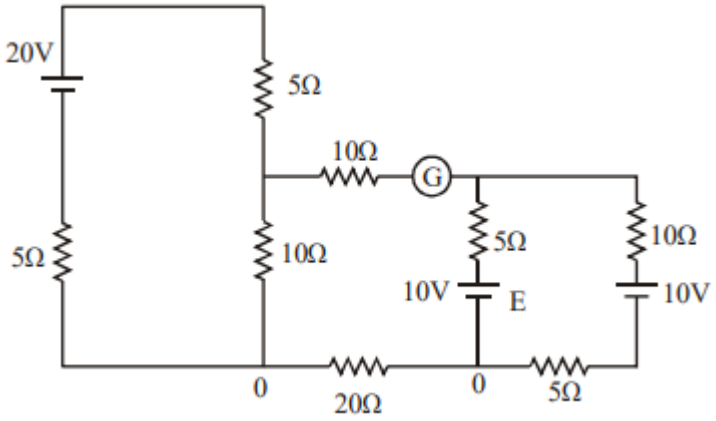
**(B)** Multimeter shows NO deflection in both cases i.e. before and after reversing the probes if the chosen component is capacitor

**(C)** Multimeter shows a deflection, accompanied by a splash of light out of connected component in one direction and NO deflection on reversing the probes if the chosen component is LED

**(D)** Multimeter shows NO deflection in both cases i.e. before and after reversing the probes is the chsosen component is metal wire.

**Q.2**

What should be value of $E$ for which galvanometer shows no deflection -




**(A)** 10 V

**(B)** 5 V

**(C)** 15 V

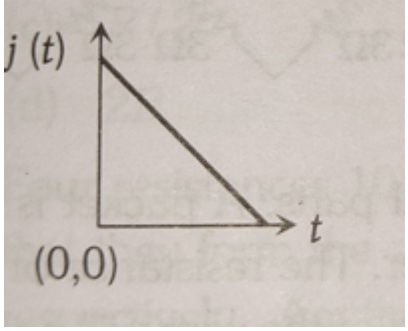
**(D)** 20 V

**Q.3**

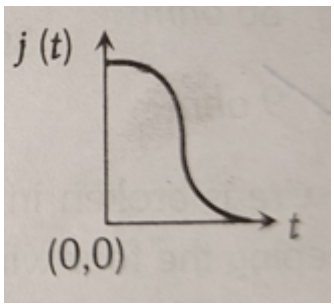
An infinite line charge of uniform electric charge density $\lambda$ lies along the axis of an electrically conducting infinite cylindrical shell of radius $R$. At time $t = 0$, the space inside the cylinder is filled with a material of permittivity $\varepsilon$ and electrical conductivity $\sigma$. The electrical conduction in the material follows Ohm's law. Which one of the following graphs best describes the subsequency variation of the magnitude of current density $f(t)$ at any point in the material

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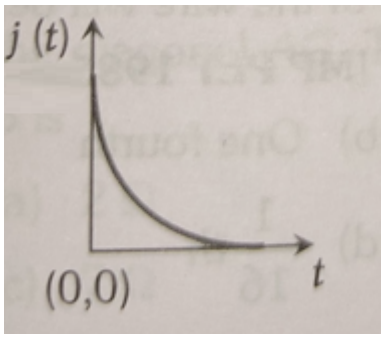
**(A)**



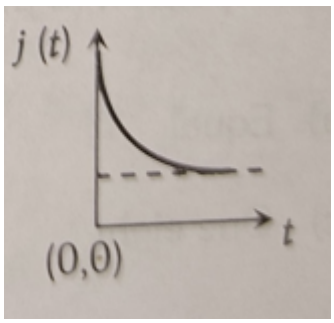

**(B)**




**(C)**

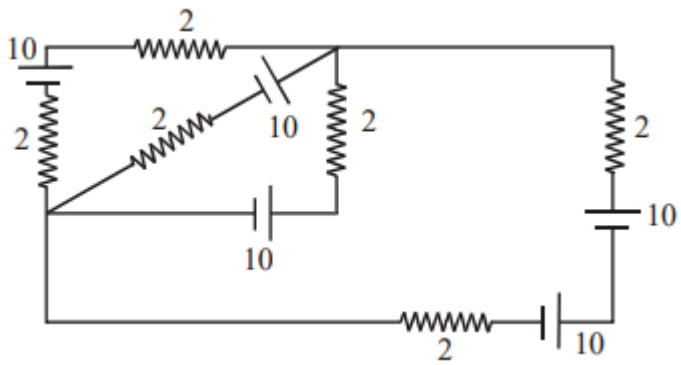



**(D)**




**Q.4**

All batteries are having emf 10 volt and internal resistance negligible. All resistors are in ohms. Calculate the current in the right most $2\Omega$ resistor.




**(A)** $\frac{25}{12}$ A

**(B)** $\frac{25}{6}$ A

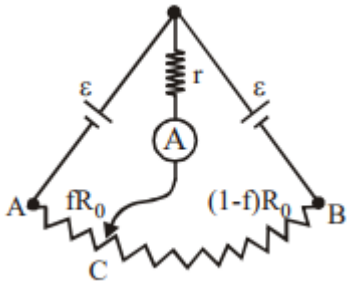
**(C)** $\frac{12}{25}$ A

**(D)** $\frac{6}{25}$ A

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**Q.5**

Figure shows a wire of uniform resistance between $\mathrm{A}$ and $\mathrm{B}$ with a total resistance of $\mathrm{R}_0$. The contact $\mathrm{C}$ can divide the wire into resistances $fR_0$ and $(1-f)R_0$. The batteries are identical and have zero internal resistances. The value of $f$ for which ammeter reads maximum is:




**(A)** 0

**(B)** 1

**(C)** 0 or 1

**(D)** cannot be found

**Q.6**

A steady current I flows through a wire of radius $r$, length $L$ and resistivity $\rho$. The current produces heat in the wire. The rate of heat loss in a wire is proportional its surface area. The steady temperature of the wire is independent of

**(A)** $L$

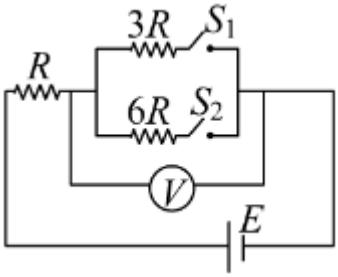
**(B)** $r$

**(C)** $\cdot I$

**(D)** $\rho$

**Q.7**

In the circuit shown in the figure, reading of voltmeter is $V_1$ when only $S_1$ is closed, reading of voltmeter is $V_2$ when only $S_2$ is closed and reading of voltmeter is $V_3$ when both $S_1$ and $S_2$ are closed. Then:




**(A)** $V_3 > V_2 > V_1$

**(B)** $V_2 > V_1 > V_3$

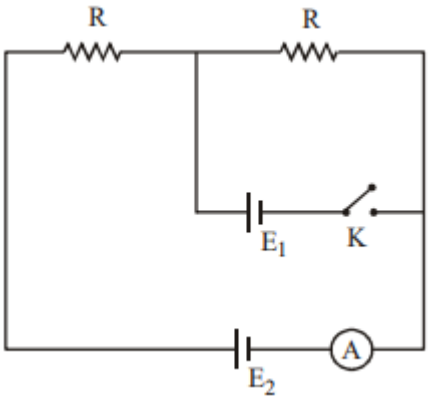
**(C)** $V_3 > V_1 > V_2$

**(D)** $V_1 > V_2 > V_3$

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**Q.8**

In the given circuit, when key $\mathrm{K}$ is open, reading of ideal ammeter is $\mathrm{I}$. Now key $\mathrm{K}$ is closed, then choose correct statement(s)-




**(A)** If $\mathrm{E_1} < \mathrm{IR}$, reading of ammeter is less than $\mathrm{I}$

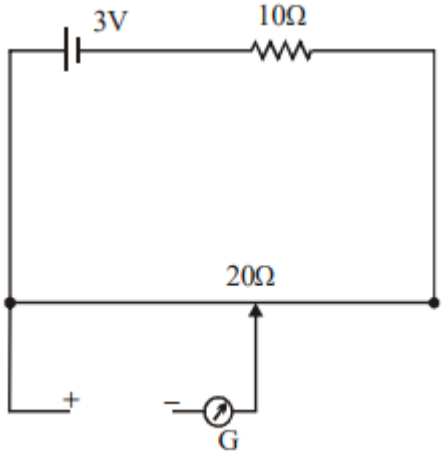
**(B)** If $\mathrm{IR} < \mathrm{E_1} < 2\mathrm{IR}$, reading of ammeter is greater than $\mathrm{I}$

**(C)** If $\mathrm{E_1} = 2\mathrm{IR}$ reading of ammeter is zero

**(D)** If $E_1 = 2IR$ reading of ammeter is 1

**Q.9**

A $10\ \mathrm{m}$ long potentiometer wire has a resistance of $20\mathrm{ohm}$. It is connected in series with a battery of emf 3V and a resistance of $10\Omega$. The internal resistance of cell is negligible. If the length can be read accurately upto $1\ \mathrm{mm}$, the potentiometer can read voltage.




**(A)** upto minimum of $0.2\mathrm{mV}$

**(B)** with an accuracy of $0.2\mathrm{mV}$

**(C)** with an accuracy of $0.1\mathrm{mV}$

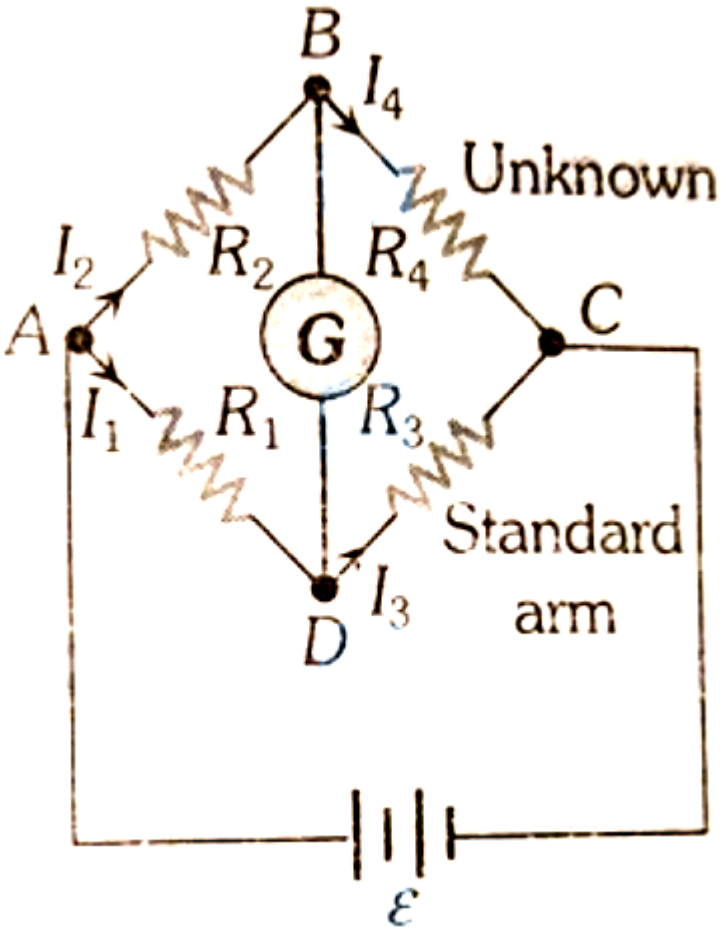
**(D)** upto maximum of 2V

**Q.10**

The measurement of an unknown resistance $R$ is to be carried out using Wheatstone's bridge as given in the figure below. Two students perform an experiment in two ways. The first students takes $R_2 = 10\Omega$ and $R_1 = 5\Omega$. The other student takes $R_2 = 1000\Omega$ and $R_1 = 500\Omega$. In the standard arm, both take $R_3 = 5\Omega$. Both find $R =$

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$\frac{R_2}{R_1}$, $R_3 = 10\Omega$ within errors




**(A)** The errors of measurement of the two students are the same

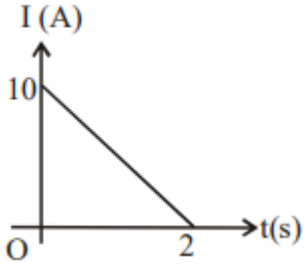
**(B)** Errors of measurement do depend on the accuracy with which $R_2$ and $R_1$ can be measured

**(C)** If the student uses large values of $R_2$ and $R_1$ the currents through the arms will be feeble. This will make determination of null point accurately more difficult

**(D)** Wheatstone bridge is a very accurate instrument and has no error of measurement

**Q.11**

A variable current flows through $1\Omega$ resistor for 2 seconds. Time dependence of the current is shown in the graph




**(A)** Total charge flown through the resistor is $10\text{C}$.

**(B)** Average current through the resistor is $5\ \text{A}$

**(C)** Total heat produced in the resistor is $50\ \text{J}$

**(D)** Maximum power during the flow of current is $100\ \text{W}$

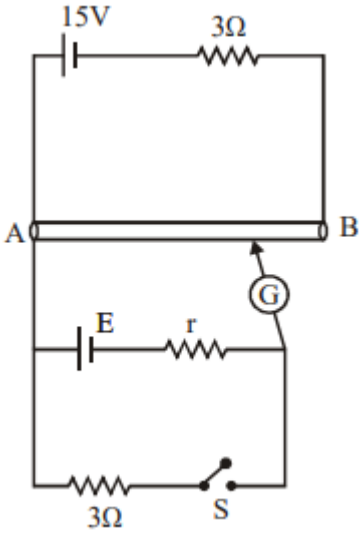
**Q.12**

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**Passage**

A wire $AB$ (of length $1\ \mathrm{m}$, area of cross section $\pi \mathrm{m}^2$ ) is used in potentiometer experiment to calculate emf and internal resistance (r) of battery. The emf and internal resistance of driving battery are $15\ \mathrm{V}$ and $3\Omega$ respectively. The resistivity of wire $\mathrm{AB}$ varies as $\rho = \rho_0 \mathrm{x}$ (where $\mathrm{x}$ is distance from $\mathrm{A}$ in meters and $\rho_0 = 24\pi\Omega$ ).

The distance of null point from $A$ is obtained at $\sqrt{\frac{2}{3}}m$ when switch $S$ is open and at $\frac{1}{\sqrt{2}}\ \mathrm{m}$ when switch is closed.




**Question**

The resistance of whole wire $\mathrm{AB}$ is -

**(A)** $6\Omega$

**(B)** $12\Omega$

**(C)** $18\Omega$

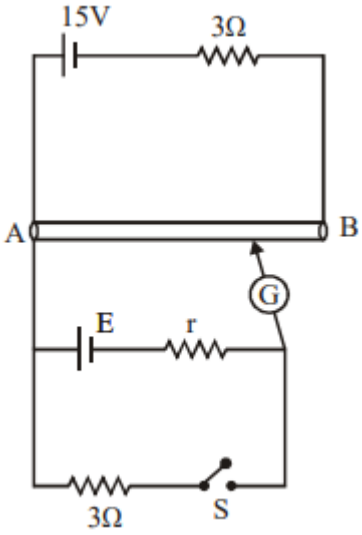
**(D)** $24\Omega$

**Q.13**

**Passage**

A wire $AB$ (of length $1\ \mathrm{m}$, area of cross section $\pi \mathrm{m}^2$ ) is used in potentiometer experiment to calculate emf and internal resistance (r) of battery. The emf and internal resistance of driving battery are $15\ \mathrm{V}$ and $3\Omega$ respectively. The resistivity of wire $\mathrm{AB}$ varies as $\rho = \rho_0 \mathrm{x}$ (where $\mathrm{x}$ is distance from $\mathrm{A}$ in meters and $\rho_0 = 24\pi\Omega$ ).

The distance of null point from $A$ is obtained at $\sqrt{\frac{2}{3}}m$ when switch $S$ is open and at $\frac{1}{\sqrt{2}}\ \mathrm{m}$ when switch is closed.




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**Question**

The current through $15\ \mathrm{V}$ battery -

**(A)** is $1\ \mathrm{A}$ only when switch $\mathrm{S}$ is closed

**(B)** is 1 A only when switch $S$ is open

**(C)** is $1\ \mathrm{A}$ in both cases

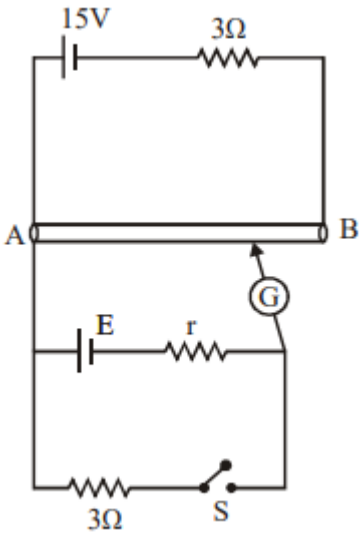
**(D)** cannot be calculated

**Q.14**

**Passage**

A wire $AB$ (of length $1\ \mathrm{m}$, area of cross section $\pi \mathrm{m}^2$ ) is used in potentiometer experiment to calculate emf and internal resistance (r) of battery. The emf and internal resistance of driving battery are $15\ \mathrm{V}$ and $3\Omega$ respectively. The resistivity of wire $\mathrm{AB}$ varies as $\rho = \rho_0 \mathrm{x}$ (where $\mathrm{x}$ is distance from $\mathrm{A}$ in meters and $\rho_0 = 24\pi\Omega$ ).

The distance of null point from $A$ is obtained at $\sqrt{\frac{2}{3}}m$ when switch $S$ is open and at $\frac{1}{\sqrt{2}}\ \mathrm{m}$ when switch is closed.




**Question**

The emf (E) of battery is -

**(A)** $4\ \mathrm{V}$

**(B)** $6\ \mathrm{V}$

**(C)** $8\ \mathrm{V}$

**(D)** $10\ \mathrm{V}$

**Q.15**

**Passage**

Resistance value of an unknown resistor is calculated using the formula $R = \frac{V}{I}$ where $V$ and $I$ be the readings of the voltmeter and the ammeter respectively. Consider the circuits below. The internal resistances of the voltmeter and

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the ammeter ($R_V$ and $R_G$ respectively) are finite and nonzero.

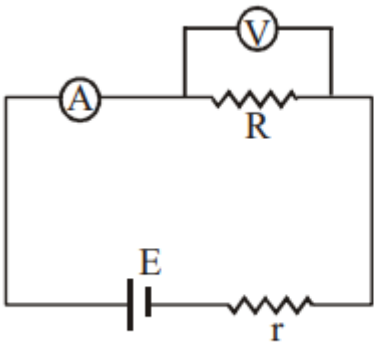


Figure (A)

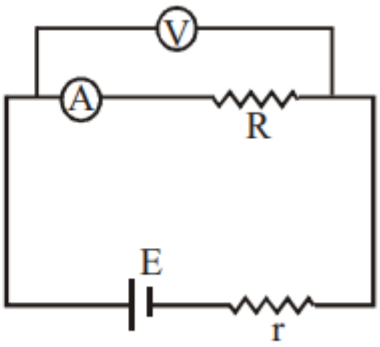


Figure (B)

Let $R_A$ and $R_B$ be the calculated values in the two cases $A$ and $B$ respectively.

**Question**

The relation between $\mathrm{R_A}$ and the actual value $\mathrm{R}$ is -

**(A)** $R > R_A$

**(B)** $\mathrm{R < R_A}$

**(C)** $\mathrm{R = R_A}$

**(D)** dependent upon $E$ and $r$

**Q.16**

**Passage**

Resistance value of an unknown resistor is calculated using the formula $R = \frac{V}{I}$ where $V$ and $I$ be the readings of the voltmeter and the ammeter respectively. Consider the circuits below. The internal resistances of the voltmeter and the ammeter ($R_V$ and $R_G$ respectively) are finite and nonzero.

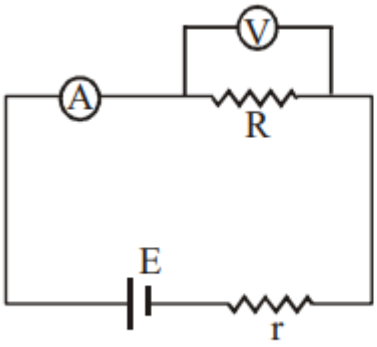


Figure (A)

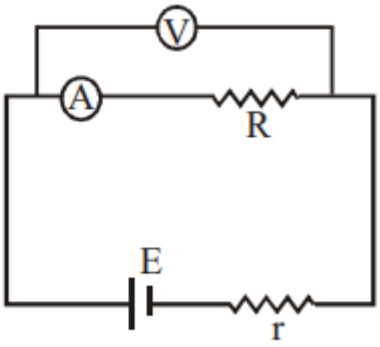


Figure (B)

Let $R_A$ and $R_B$ be the calculated values in the two cases $A$ and $B$ respectively.

**Question**

The relation between $R_B$ and the actual value $R$ is $-$

**(A)** $\mathrm{R < R_B}$

**(B)** $\mathrm{R > R_B}$

**(C)** $\mathrm{R = R_B}$

**(D)** dependent upon $\mathrm{E}$ and $\mathrm{r}$

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**Q.17**

**Passage**

Resistance value of an unknown resistor is calculated using the formula $R = \frac{V}{I}$ where $V$ and $I$ be the readings of the voltmeter and the ammeter respectively. Consider the circuits below. The internal resistances of the voltmeter and the ammeter ($R_V$ and $R_G$ respectively) are finite and nonzero.

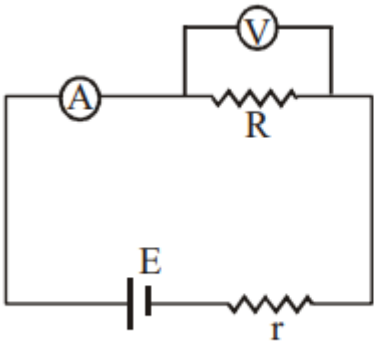


Figure (A)

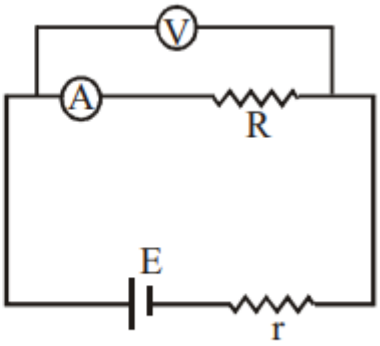


Figure (B)

Let $R_A$ and $R_B$ be the calculated values in the two cases $A$ and $B$ respectively.

**Question**

If the resistance of voltmeter is $\mathrm{R_V} = 1\mathrm{k}\Omega$ and that of ammeter is $\mathrm{R_G} = 1\Omega$, the magnitude of the percentage error in the measurement of $R$ (the value of $R$ is nearly $10\Omega$ ) is -

**(A)** zero in both cases

**(B)** non zero but equal in both cases

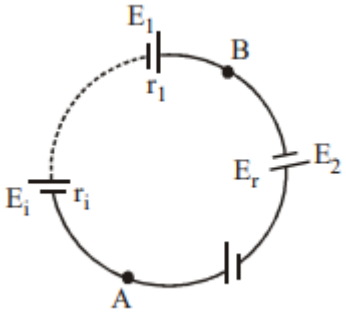
**(C)** more in circuit A

**(D)** more in circuit B

**Q.18**

n cells are connected in a closed loop. Emf of the cells are $1\ \mathrm{V}, 2\ \mathrm{V}, 3\ \mathrm{V}, \ldots\ldots \mathrm{nV}$. Internal resistance of the cells are $0.5\Omega, 1.0\Omega, 1.5\Omega, 2\Omega, \ldots\ldots\ldots \frac{\mathrm{n}}{2}\Omega$.

Based on above facts, match column I with column II.




| Column I | Column II |
|---|---|
| (A) Current through each cell | (p) zero |
| (B) Terminal voltage of each cell | (q) equal |
| (C) Potential difference between any two points A and B on the circuit | (r) unequal |
| (D) Power supplied by each cell | (s) non-zero |

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**(A)** $(A)-s(B)-p,r,s(C)-q,s(D)-p,r$

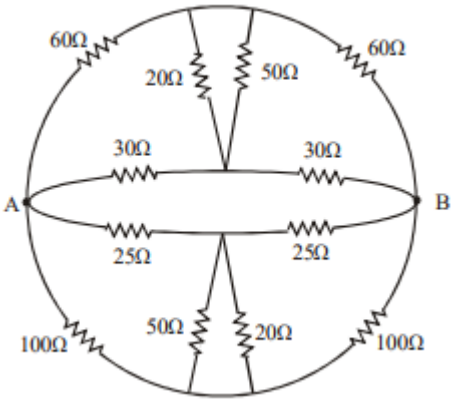
**(B)** $(A)-q,s(B)-p,q(C)-p,q(D)-r,s$

**(C)** $(A)-p,r(B)-q(C)-q,s(D)-p$

**(D)** $(A)-p,s(B)-p,r,s(C)-q,s(D)-q,r$

**Q.19**

Equivalent resistance is $10\mathrm{x}\Omega$ between points $\mathrm{A}$ and $\mathrm{B}$. Find the value of $\mathrm{x}$. Values of resistance are as shown in figure.




**Q.20**

The area of cross section, length, and density of a piece of metal of atomic weight 60 are $10^{-6}\ \mathrm{m}^2, 1\ \mathrm{m}$, and $5\times 10^3 \mathrm{kgm}^{-3}$, respectively. If every atom contributes one free electron, find the drift velocity (in $\mathrm{mms}^{-1}$ ) of electrons in the metal when a current of $16\ \mathrm{A}$ passes through it. Avogadro's number is $N_A=6\times 10^{23}/\mathrm{mol}$ and charge on an electron is $e=1.6\times 10^{-19}\mathrm{C}$.

## Answers & Solutions

**Q.1 Answer:**

Multimeter shows a deflection, accompanied by a splash of light out of connected component in one direction and NO deflection on reversing the probes if the chosen component is LED

**Solution:**

**The multimeter does not deflect before and after reversing the probes in the case of a capacitor.**

**-In the case of a resistor, the multimeter shows an equal deflection before and after reversing the probes in both cases.**

**- When inverting the probes on a metal wire, the multimeter does not show any deflection before and after.**

**- When the probes are reversed, the multimeter indicates deflection in one direction, accompanied by a splash of light from the connected component, but no deflection when the probes are reversed in the case of an LED.**

**Q.2 Answer:**

10 V

**Solution:**

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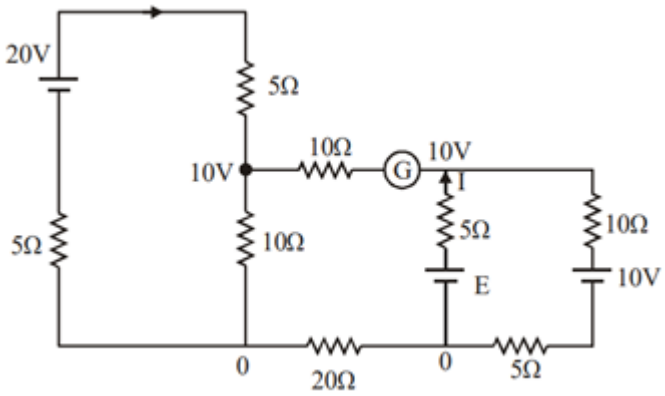
$\frac{E-10}{20} = I$

$10 = E - \frac{E-10}{20} \times 5$

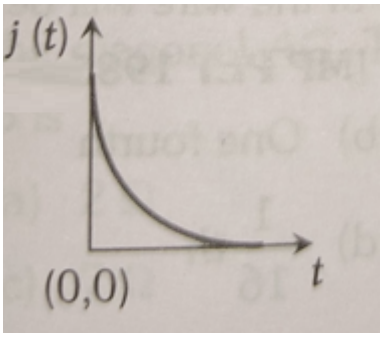
$40 = 4E - E + 10$

$30 = 3E$

$E = 10\ \text{V}$




**Q.3 Answer:**




**Solution:**

**For infinite line**

$$E = \frac{\lambda}{2\pi\varepsilon r}$$

$$\Rightarrow dV = \frac{-\lambda}{2\pi\varepsilon r} dr$$

**Current through an elemental shell;**

$$I = \frac{|dV|}{dR} = \frac{\frac{\lambda}{2\pi\varepsilon r} dr}{\frac{1}{\sigma} \times \frac{dr}{2\pi r l}} = \frac{\lambda\sigma l}{\varepsilon}$$

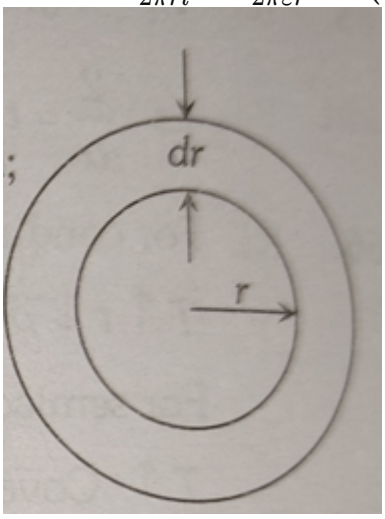
**This current is radially outwards so;**

$$\frac{d}{dt}(\lambda l) = \frac{-\lambda\sigma l}{\varepsilon} \Rightarrow \frac{d\lambda}{\lambda} = -\left(\frac{\sigma}{\varepsilon}\right) dt$$

$$\Rightarrow \lambda = \lambda_0 e^{-(\sigma/\varepsilon)t}$$

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So, $j = \frac{I}{2\pi rl} = \frac{\lambda\sigma}{2\pi\varepsilon r} = \left(\frac{\lambda_0\sigma}{2\pi\varepsilon r}\right) e^{-(\sigma/\varepsilon)t}$

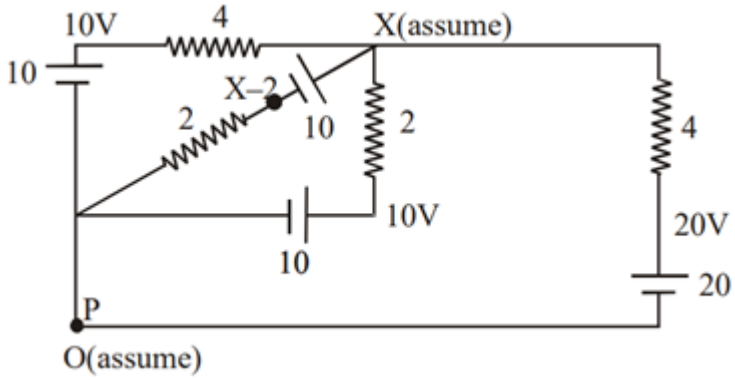



**Q.4 Answer:**

$\frac{25}{12}$ A

**Solution:**

**The simplified circuit is**




**We have to find I.**

**Let potential of point $P$ be $0$.**

**Potential at other points are shown in the figure apply Kirchoff's current law at $x_x$.**

$$\frac{x-10}{4}+\frac{x-10}{2}+\frac{x-20}{4}+\frac{(x-10)-0}{2}=0$$

$$\Rightarrow x-10+2x-20+x-20+2x-20=0 \Rightarrow 6x=70 \Rightarrow x=\frac{35}{3}\text{ volt}$$

$$\therefore \quad \mathrm{I}=\frac{20-\frac{35}{3}}{4}=\frac{25}{12}\ \mathrm{A}$$

$\frac{x-10}{4}+\frac{x-10}{2}+\frac{x-20}{4}+\frac{(x-10)}{2}=0$

**or** $x=\frac{35}{3}V$

$I=\frac{20-\frac{35}{3}}{4}=\frac{25}{12}$ A

**Q.5 Answer:**

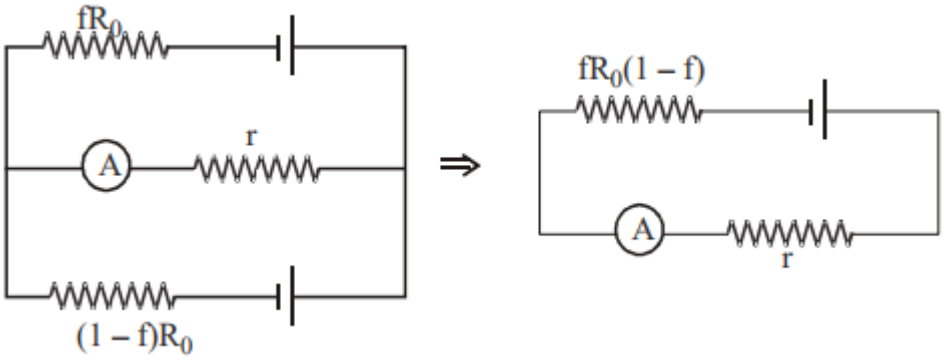
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0 or 1

**Solution:**

$i = \frac{\varepsilon}{r + fR_0(1-f)}$

$\mathrm{i} \to \max$ **if** $\mathrm{fR_0(1-f) = 0 \Rightarrow f = 0}$ **or 1**




**Q.6 Answer:**

$L$

**Solution:**

$R = \frac{\rho l}{A} = \frac{\rho l}{\pi r^2}$

**Now,** $I^2R = K2\pi rl\frac{dT}{dt}$ $(kx =$ **arbitrary constant)** $\Rightarrow \frac{I^2\rho l}{\pi r^2} = K2\pi rl\frac{dT}{dt} \Rightarrow \frac{dT}{dt} = \frac{I^2\rho}{K2\pi^2 r^3}$

**It is independent of length** $(l)$**.**

**Q.7 Answer:**

$V_2 > V_1 > V_3$

**Solution:**

**In series P.D.** $\propto R$

**When only** $S_1$ **is closed,** $V_1 = \frac{3}{4}E = 0.75E$

**When only** $S_2$ **is closed,** $V_2 = \frac{6}{7}E = 0.86E$

**And when** $S_1$ **and** $S_2$ **are closed, combined resistance of** $6R$ **and** $3R$ **is** $2R$**.**

$\therefore \quad V_3 = \frac{2}{3}E = 0.67E \quad \therefore \quad V_2 > V_1 > V_3$

**Q.8 Answer:**

If $\mathrm{E_1 = 2IR}$ reading of ammeter is zero

**Solution:**

$\mathrm{E_2 = 2IR}$

$$\mathrm{I' = \frac{E_2 - E_1}{R} = 2I - \frac{E_1}{R}}$$
$$\mathrm{If\ E_1 < IR \Rightarrow I' > I}$$
$$\mathrm{If\ IR < E_1 < 2IR \Rightarrow I' < I}$$
$$\mathrm{If\ E_1 = 2IR \Rightarrow I' = 0}$$

**Q.9 Answer:**

with an accuracy of $0.2\mathrm{mV}$

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**Solution:**

**Current in wire** $= \frac{3}{20} = 0.1$ A

$$\mathrm{V} = \mathrm{IR} = 2 \text{ V}$$

**maximum voltage can read** $= 2$ V

$$\lambda = \frac{\mathrm{V}}{\ell} = \frac{2}{10 \times 10^3} = 0.2\mathrm{mV/mm}$$

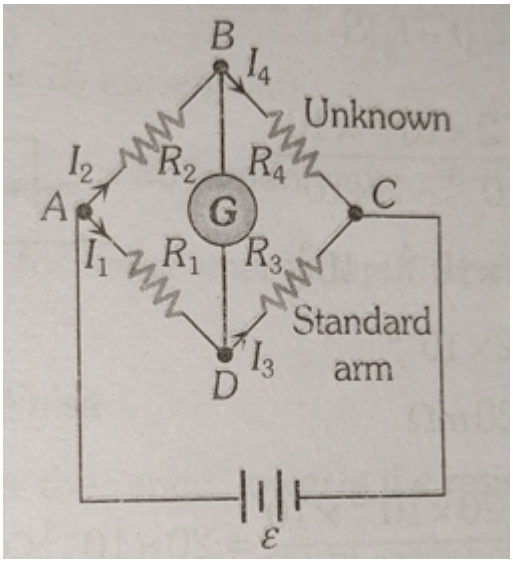
**Least account of scale** $= 1$ mm, **accuracy** $= 0.2\mathrm{mV}$

**Q.10 Answer:**

Errors of measurement do depend on the accuracy with which $R_2$ and $R_1$ can be measured

**Solution:**

**Given, for first student,** $R_2 = 10\Omega, R_1 = 5\Omega, R_3 = 5\Omega$ **For second student,** $R_1 = 500\Omega, R_3 = 5\Omega$
**Now, according to Wheatstone bridge rule,** $\frac{R_2}{R_4} = \frac{R_1}{R_3} \Rightarrow R_4 = R_3 \times \frac{R_2}{R_1}$




**Now, putting all the values in Eq. (i), we get** $R = 10\Omega$ **for both students. Thus, we can analyse that the Wheatstone bridge is most sensitive and accurate if resistances are of same value.**
**Thus, the errors of measurement of the two students depend on the accuracy and sensitivity of the bridge. which in turn depends on the accuracy with which** $R_2$ **and** $R_1$ **can be measured.**
**When** $R_2$ **and** $R_1$ **are larger, the currents through the When of bridge is very weak. This can make the determination of null point accurately more difficult.**

**Q.11 Answer:**

Total charge flown through the resistor is $10\mathrm{C}$.

**Solution:**

**Total charge** $= \int \mathrm{Idt} =$ **Area under the curve** $= 10\mathrm{C}$
**Average current** $= \frac{\int I dt}{\int dt} = 5A$

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**Total heat produced** $= \int I^2 R dt = \int_0^2 (-5t + 10)^2 \cdot 1 \cdot dt = \frac{200}{3}$ J

**Maximum power** $= I^2 R$ **when** $I$ **is maximum current** $= 100 \times 1 = 100$ W

**Q.12 Answer:**

$12\Omega$

**Solution:**

**Resistance of wire** $AB$ **is** $R_{AB} = \left(\frac{\rho_0 \ell}{2}\right) \frac{\ell}{A} = \frac{24\pi}{2\pi} = 12\Omega \quad \left\{R = \int_0^1 \frac{\rho dx}{A}\right\}$

**Current in wire** $AB$ **is** $I = \frac{15}{12+3} = 1$ A

**Q.13 Answer:**

is $1$ A in both cases

**Solution:**

**Resistance of wire** $AB$ **is** $R_{AB} = \left(\frac{\rho_0 \ell}{2}\right) \frac{\ell}{A} = \frac{24\pi}{2\pi} = 12\Omega \quad \left\{R = \int_0^1 \frac{\rho dx}{A}\right\}$ **Current in wire** $AB$ **is** $I = \frac{15}{12+3} = 1$ A

$R_{AC} = \left(\frac{\rho_0 x}{2}\right)\left(\frac{x}{A}\right) = \frac{\rho_0 x^2}{2A} = \frac{24\pi \frac{2}{3}}{2\pi} = 8\Omega$

**When switch is open, null point at** $C(AC = x)$

$R_{AD} = \left(\frac{\rho_0 x}{2}\right)\left(\frac{x}{A}\right) = \frac{\rho_0 x^2}{2A} = \frac{24\pi \frac{1}{2}}{2\pi} = 6\Omega$

**EMF,** $E = 1 \times 8 = 8$ V

**When switch closed null point at** $D(AD = x)$

**Q.14 Answer:**

$8$ V

**Solution:**

**Resistance of wire** $AB$ **is** $R_{AB} = \left(\frac{\rho_0 \ell}{2}\right) \frac{\ell}{A} = \frac{24\pi}{2\pi} = 12\Omega \quad \left\{R = \int_0^1 \frac{\rho dx}{A}\right\}$ **Current in wire** $AB$ **is** $I = \frac{15}{12+3} = 1$ A

$R_{AC} = \left(\frac{\rho_0 x}{2}\right)\left(\frac{x}{A}\right) = \frac{\rho_0 x^2}{2A} = \frac{24\pi \frac{2}{3}}{2\pi} = 8\Omega$

**When switch is open, null point at** $C(AC = x)$

$R_{AD} = \left(\frac{\rho_0 x}{2}\right)\left(\frac{x}{A}\right) = \frac{\rho_0 x^2}{2A} = \frac{24\pi \frac{1}{2}}{2\pi} = 6\Omega$

**EMF,** $E = 1 \times 8 = 8$ V

**When switch closed null point at** $D(AD = x)$

**Q.15 Answer:**

$R > R_A$

**Solution:**

$R_A = \frac{R \cdot R_V}{R + R_V} < R$

**Q.16 Answer:**

$R < R_B$

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**Solution:**

$$R_B = R + R_G > R$$

**Q.17 Answer:**

more in circuit B

**Solution:**

$\%$ **error in case** $A$ : $\frac{R_A - R}{R} \times 100 = \left(\frac{R_V}{R+R_V} - 1\right) \times 100 = \frac{-R}{R+R_V} \times 100 \approx -1\%$

$\%$ **error in case** $\mathrm{B}$ : $\frac{\mathrm{R_B - R}}{\mathrm{R}} \times 100 = \frac{\mathrm{R_G}}{\mathrm{R}} \times 100 \approx 10\%$

**Hence percentage error in circuit B is more than that in A.**

**Q.18 Answer:**

$(A) - q, s (B) - p, q (C) - p, q (D) - r, s$

**Solution:**

**(A)-q s, (B)-p q, (C)-p q, (D)-r s**

$\mathrm{i} = \frac{\mathrm{E_1 + E_2 + \ldots . E_n}}{\mathrm{r_1 + r_2 + \ldots . r_n}} = \frac{1+2+3+\ldots \mathrm{n}}{0.5+1.0+1.5+\ldots . \mathrm{n}/2} = \frac{\frac{\mathrm{n}}{2}(1+\mathrm{n})}{\frac{\mathrm{n}}{2}\left(\frac{1}{2}+\frac{\mathrm{n}}{2}\right)} = 2$

**Terminal voltage (cell 1)** $= \mathrm{E_1 - ir_1} = 1 - 2 \times 0.5 = 0$

**Terminal voltage** $($ **cell 2**$) = \mathrm{E_2 - ir_2} = 2 - 2 \times 1 = 0$

**Power supplied by cell** $1 = i^2 r_1$

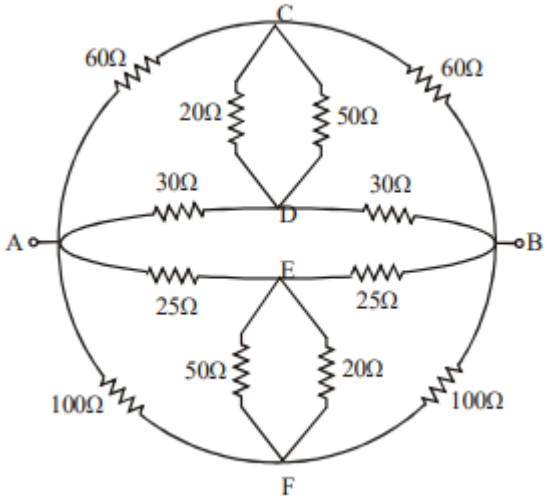
**Power supplied by cell** $2 = i^2 r_2 \ldots \ldots$

**Q.19 Answer:**

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**Solution:**

$\mathrm{ACDE}$ **and** $\mathrm{AEBF}$ **are two balanced Wheatstone bridges each with equivalent resistance** $40\Omega$**, therefore total resistance between** $\mathrm{AB}$ **is** $20\Omega$**.**




**Q.20 Answer:**

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**Solution:**

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**As according to Avogadro's hypothesis,** $\frac{N}{N_A}=\frac{m}{M}$ **and** $n=\frac{N}{V}=N_A\frac{m}{VM}=\frac{N_Ad}{M}\quad\left[\text{ as } d=\frac{m}{V}\right]$

**Electron density** $n=\frac{6\times10^{23}\times\left(5\times10^3\right)}{\left(60\times10^{-3}\right)}=5\times10^{28}\ \mathrm{m}^{-3}$ **Now as each atom contributes one electron, the number of electrons per unit volume is** $1\times n=1\times5\times10^{28}=5\times10^{28}\ \mathrm{m}^{-3}$

**Further** $J=\frac{i}{A}=\frac{16}{10^{-6}}=16\times10^6\mathrm{Am}^{-2}$

$J=nev_d$ **or** $v_d=\frac{J}{ne}=2\times10^{-3}=2\mathrm{mms}^{-1}$.

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